Beyond Confidence: Learning to Abstain in Widlife Classification Under Distribution Shift

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Abstract

Automated wildlife monitoring relies on image classifiers to identify animals in camera-trap photographs. However, changes in backgrounds, lighting, and camera placement can make predictions unreliable at locations absent from training. This project investigates whether selective classification—allowing a model to abstain from answering—can improve prediction reliability under these distribution shifts. Using the iWildCam-WILDS dataset and a fixed wildlife classifier, we will reproduce and adapt Heng et al.'s likelihood-ratio-based selective classification method as our research baseline. We will investigate whether learning to combine its reliability signals with prediction consistency across mild image augmentations improves the identification of classification errors. A further experiment will assess whether balancing camera-location contributions during reliability-estimator training improves generalization to unseen sites. Methods will be evaluated using error rates at matched coverage and risk–coverage curves, with species- and location-level analysis to reveal uneven rejection patterns. Ablation experiments will isolate each component's contribution. Our objective is to reduce errors among accepted predictions without simply rejecting more images, supporting wildlife monitoring systems that flag unreliable predictions for human review.

1 Introduction

Camera traps help ecologists monitor wildlife by collecting large numbers of animal photographs. Automated classification can reduce the effort required to label these images, but models deployed at new camera locations encounter changes in lighting, vegetation, backgrounds, and viewing angles. These changes can produce incorrect predictions that nevertheless receive high confidence. Identifying such predictions and flagging them for human review could improve the reliability of automated wildlife monitoring.

Our objective is to investigate whether a learned reliability estimator can improve selective classification under camera-location distribution shift. Rather than changing the underlying classifier, we will study how effectively different methods distinguish its correct predictions from its mistakes. Specifically, we will:

- **Establish a published research baseline:** reproduce Heng et al.'s likelihood-ratio-based selective classification method and adapt it to iWildCam-WILDS
- **Evaluate augmentation consistency:** test whether prediction stability across mild image transformations provides additional information for identifying errors
- **Evaluate performance and limitations:** compare error rates at matched coverage, including 80% and 90%, and risk–coverage curves; use ablations and species- and location-level analysis to identify failure patterns

33rd Conference on Neural Information Processing Systems (NeurIPS 2019), Vancouver, Canada.

2 Review of related literature

Wildlife classification under distribution shift Koh et al. [2021] introduce WILDS, a benchmark of naturally occurring distribution shifts. Its iWildCam-WILDS task evaluates animal classification across camera-trap locations, where backgrounds, illumination, viewpoints, and species frequencies can differ. This provides a directly relevant setting for evaluating whether an abstention mechanism generalizes to unseen deployment sites

Likelihood-ratio-based selective classification Heng and Soh [2026] formulate selective classification through the Neyman–Pearson framework, relating optimal selection to a likelihood ratio distinguishing correct from incorrect predictions. They develop feature-distance scores, Δ-MDS and Δ-KNN, and combine them with logit-margin information. Their experiments demonstrate improvements over competing selectors on the evaluated distribution-shift benchmarks, but do not include iWildCam. We will reproduce a published score combination and adapt it to wildlife classification as our primary baseline. Our learned combination is an empirical extension; it does not automatically inherit the paper's theoretical optimality guarantees

3 Methodology

3.1 Model Description

We will use CLIP ViT-B/16 fine-tuned using FLYP, following the official iWildCam training command, as our base classifier. After fine-tuning, it will remain fixed across selector comparisons. Our abstention baseline will adapt Heng and Soh's feature-distance and logit-margin score combination. We will investigate learned signal combinations, augmentation-consistency features, and location-balanced training of the reliability estimator.

3.2 Dataset

We will use iWildCam-WILDS, which contains labeled camera-trap images, camera-location metadata, and predefined familiar- and unseen-location evaluation splits. Images will follow the preprocessing used by FLYP's CLIP ViT-B/16 implementation. We will reserve a subset of training locations, excluded from classifier fitting, to train the reliability estimator. Official validation data will guide hyperparameter and threshold selection, while test data will remain reserved for final evaluation.

3.3 Evaluation Metric

Our primary measures will be coverage, the fraction of images receiving an accepted prediction, and selective risk, the error rate among accepted predictions.These measures assess whether a method identifies unreliable predictions effectively while retaining useful automated coverage

We will compare selectors using risk–coverage curves, area under these curves (AURC), and selective risk at 80% and 90% coverage. Matched-coverage comparisons will use the same fixed classifier, isolating improvements in error identification. For thresholds selected on validation data, we will report both achieved test coverage and selective risk. Overall accuracy and macro-F1 will characterize the classifier, while species- and location-level risk and coverage will reveal uneven rejection patterns.

3.4 Experimental Depth and Iteration

We will establish FLYP ViT-B/16 classifier performance and implement Heng and Soh's selector as our primary baseline, with maximum softmax probability and logit margin as additional comparisons. We will then compare the published signal combination with logistic regression and a small neural reliability estimator.

Ablations will assess the added value of augmentation consistency and location-balanced estimator training. We will vary augmentation type and number, record additional inference cost, and repeat selected configurations across random seeds. Failure analysis will examine confidently incorrect predictions and correct predictions unnecessarily rejected.

4 Baseline and Extension

4.1 Baseline selection and Evaluation

We will use CLIP ViT-B/16 fine-tuned with FLYP as the fixed classifier and adapt Heng and Soh's likelihood-ratio-inspired feature-distance and logit-margin combination as our primary abstention baseline. Maximum softmax probability and logit margin will provide additional comparisons.

Evaluation will use AURC and selective risk at matched coverage on unseen camera locations. Validation-selected thresholds will also be evaluated by reporting achieved test coverage and risk.

4.2 Proposed Extensions and Experiments

We will investigate three extensions:

- **Learned signal combination:** compare the baseline's prescribed combination with logistic regression and a small neural reliability estimator
- **Augmentation consistency:** add prediction-stability features from mild, class-preserving transformations to test their complementary value.
- **Location-balanced training:** give camera sites more equal influence when training the reliability estimator to assess generalization to unseen locations

Ablations will isolate each extension's contribution using the same fixed classifier. We will also measure augmentation-related inference cost and examine species- and location-level rejection patterns

5 Anticipated Results

We hypothesize that learned signal combinations and augmentation consistency will reduce selective risk relative to Heng and Soh's baseline at matched coverage. Location-balanced training may improve generalization across unseen camera sites, but gains are not guaranteed.

We will report risk–coverage curves, AURC, and selective risk at 80% and 90% coverage, alongside achieved test coverage and risk for validation-selected thresholds. Ablations and species- and location-level analysis will identify which extensions help and where they remain unreliable

6 Discussion

Potential challenges include class imbalance, small or partially visible animals, and consistently incorrect predictions. Learned reliability scores may overfit training locations, while augmentations may obscure useful information and increase inference cost. We will assess whether improvements justify this cost and avoid disproportionate rejection of particular species or camera sites

7 Planned Work

Before the midterm report, we will

- Prepare iWildCam-WILDS and establish leakage-free data splits.
- Fine-tune CLIP ViT-B/16 using FLYP
- Implement Heng and Soh's abstention baseline.
- Produce initial classification metrics and risk–coverage curves.
- Explore augmentation consistency technique if time permits

References

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